

Matematica - Clasa a 12-a

Lectii esentiale - Toate capitolele

1. Limite de functii - formule cheie

$$\lim_{x \rightarrow 0} \sin(x)/x = 1 \text{ (limita fundamentala)}$$

$$\lim_{x \rightarrow 0} (1 - \cos x)/x^2 = 1/2$$

$$\lim_{x \rightarrow \infty} (1 + 1/x)^x = e \text{ (} e \approx 2,718 \text{)}$$

$$\lim_{x \rightarrow 0} (e^x - 1)/x = 1$$

$$\lim_{x \rightarrow 0} \ln(1+x)/x = 1$$

$$\text{Regula L'Hopital: } \lim f/g = \lim f'/g' \text{ (} 0/0 \text{ sau } \infty/\infty \text{)}$$

2. Derivate - toate formulele

$$\bullet (c)' = 0$$

$$\bullet (x^n)' = nx^{n-1}$$

$$\bullet (\sqrt{x})' = 1/(2\sqrt{x})$$

$$\bullet (e^x)' = e^x$$

$$\bullet (a^x)' = a^x \ln a$$

$$\bullet (\ln x)' = 1/x$$

$$\bullet (\log_a x)' = 1/(x \ln a)$$

$$\bullet (\sin x)' = \cos x$$

$$\bullet (\cos x)' = -\sin x$$

$$\bullet (\tan x)' = 1/\cos^2 x$$

$$\bullet (\cot x)' = -1/\sin^2 x$$

$$\bullet (\arcsin x)' = 1/\sqrt{1-x^2}$$

$$\bullet (\arccos x)' = -1/\sqrt{1-x^2}$$

$$\bullet (\arctg x)' = 1/(1+x^2)$$

$$(f+g)' = f' + g'; \quad (fg)' = f'g + fg'; \quad (f/g)' = (f'g - fg')/g^2$$

$$\text{Compusa: } (f(g(x)))' = f'(g(x)) \times g'(x)$$

Clasa 12 - Studiul functiei

3. Studiul complet al unei functii

- 1. Domeniu de definitie
- 2. Paritate (para/impara/niciuna)
- 3. Asimptote (verticale, orizontale, oblice)
- 4. Derivata $f'(x)$: puncte critice, semnul lui f'
- 5. Monotonie: crescatoare / descrescatoare
- 6. Extreme locale (maxime/minime locale)
- 7. Derivata a doua $f''(x)$: convexitate, concavitate
- 8. Puncte de inflexiune ($f''=0$ si schimba semn)
- 9. Tabel de variatie
- 10. Graficul functiei

4. Asimptote - formule

Asimptota verticala $x=a$: $\lim_{x \rightarrow a} f(x) = \pm \infty$

Asimptota orizontala $y=L$: $\lim_{x \rightarrow \pm \infty} f(x) = L$

Asimptota oblica $y=mx+n$:

$$m = \lim_{x \rightarrow \infty} f(x)/x; \quad n = \lim_{x \rightarrow \infty} (f(x) - mx)$$

5. Exemple de studiu al functiei

$$f(x) = (x^2 - 1) / x = x - 1/x$$

Domeniu: $x \neq 0$

Asimptota verticala: $x=0$ ($\lim = \pm \infty$)

Asimptota oblica: $m=1, n=-1 \Rightarrow y=x$ ($la +\infty$ si $-\infty$)

$$f'(x) = 1 + 1/x^2 > 0 \text{ pt orice } x \neq 0 \Rightarrow \text{mereu crescatoare}$$

Fara extreme locale (f' nu se anuleaza)

$$f''(x) = -2/x^3: <0 \text{ pt } x>0 \text{ (concava), } >0 \text{ pt } x<0 \text{ (convexa)}$$

Clasa 12 - Integrala

6. Primitive elementare

- $\int x^n dx = x^{(n+1)}/(n+1) + C$ ($n \neq -1$)
- $\int 1/x dx = \ln|x| + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = a^x / \ln a + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int 1/\cos^2 x dx = \tan x + C$
- $\int 1/\sin^2 x dx = -\cot x + C$
- $\int 1/\sqrt{1-x^2} dx = \arcsin x + C$
- $\int 1/(1+x^2) dx = \arctg x + C$

7. Metode de integrare

Substitutia:

Setam $u=g(x)$, $du=g'(x)dx$, inlocuim in integrala

ex: $\int \sin(3x)dx$: $u=3x$, $du=3dx \Rightarrow 1/3 \int \sin u du = -\cos(3x)/3 + C$

Integrarea prin parti:

$$\int u dv = uv - \int v du$$

- Alegem u = functia care se simplifica la derivare (polinoame, $\ln$, arcfunctii)
- Alegem dv = restul (exp, trig)

ex: $\int x e^x dx$: $u=x$, $dv=e^x dx \Rightarrow x e^x - e^x + C$

Integrala definita:

$$\int [a,b] f(x)dx = F(b)-F(a) \text{ (Leibniz-Newton)}$$

Aria = $\int [a,b] |f(x)|dx$; intre curbe = $\int |f-g|dx$

Volum rotatie in jurul Ox: $V = \pi \int [a,b] f^2(x)dx$

Clasa 12 - Algebra liniara

8. Matrice

- Operatii: +, $\times$ scalar, produs $(AB)_{ij} = \sum a_{ik} b_{kj}$
- $AB \neq BA$ (comutativitate NU se aplica in general)
- $(AB)^T = B^T A^T$; $(A^{-1})^T = (A^T)^{-1}$

Matrice unitate I : $AI = IA = A$

Inversa: $A^{-1} = (1/\det A) \times \text{Adj}(A)$

9. Determinanti

$$\det(2 \times 2) = ad - bc$$

$\det(3 \times 3)$: Sarrus sau dezvoltare dupa linie/coloana

$$\det(AB) = \det(A) \det(B); \det(A^T) = \det(A)$$

$$\det(cA) = c^n \det(A) \quad (A \text{ matrice } n \times n)$$

- Schimb 2 linii: semn se schimba
- Linie nula $\Rightarrow \det = 0$; 2 linii egale $\Rightarrow \det = 0$
- $\det(A) = 0 \Leftrightarrow A$ singulara $\Rightarrow$ sistemul nu are sol. unica

10. Sisteme liniare

Cramer ($\det A \neq 0$):

$$x_i = \det(A_i) / \det(A)$$

Gauss (eliminare gaussiana):

- $[A/b] \rightarrow$ forma treapta superior
- Sistem compatibil det. $\Rightarrow$ sol. unica
- Sistem compatibil nedet. $\Rightarrow$ infinit de solutii
- Sistem incompatibil $\Rightarrow$ nicio solutie

Clasa 12 - Combinatorica si Probabilitati

11. Combinatorica

$$n! = 1 \times 2 \times \dots \times n; \quad 0! = 1$$

$$A(n, k) = n! / (n-k)! \quad (\text{aranjamente ord. conteaza})$$

$$C(n, k) = n! / (k!(n-k)!) \quad (\text{combinari ord. nu conteaza})$$

$$P(n) = n! \quad (\text{permutari})$$

$$(a+b)^n = \sum_{k=0..n} C(n, k) a^{n-k} b^k \quad (\text{Newton})$$

$$C(n, 0) + C(n, 1) + \dots + C(n, n) = 2^n$$

$$\text{Pascal: } C(n, k) + C(n, k+1) = C(n+1, k+1)$$

12. Probabilitati

$$P(A) = \text{card}(A) / \text{card}(\Omega); \quad 0 \leq P(A) \leq 1$$

$$P(A \text{ uniune } B) = P(A) + P(B) - P(A \text{ intersect } B)$$

$$P(A/B) = P(A \text{ intersect } B) / P(B) \quad (\text{conditionata})$$

$$\text{Independenta: } P(A \text{ intersect } B) = P(A) \times P(B)$$

$$\text{Prob. totala: } P(A) = \sum P(A/H_i) P(H_i)$$

$$\text{Bayes: } P(H_i/A) = P(A/H_i)P(H_i) / P(A)$$

13. Distributii de probabilitate

$$\text{Binomiala } B(n, p): P(X=k) = C(n, k) p^k (1-p)^{n-k}; \quad E=np; \quad D=np(1-p)$$

$$\text{Normala } N(\mu, \sigma^2): E=\mu; \quad D=\sigma^2$$

$$\text{Standardizare: } Z = (X-\mu)/\sigma \sim N(0, 1)$$

- Regula 68-95-99.7: $P(|X-\mu| < \sigma) = 68\%$, $< 2\sigma = 95\%$, $< 3\sigma = 99.7\%$

Clasa 12 - Trigonometrie & Geometrie

14. Trigonometrie avansata

$$\sin^2 x + \cos^2 x = 1$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\sin 2a = 2 \sin a \cos a; \cos 2a = 2\cos^2 a - 1$$

$$\text{Formule produs} \rightarrow \text{suma: } \sin a \cos b = (\sin(a+b) + \sin(a-b))/2$$

$$\text{Formule suma} \rightarrow \text{produs: } \sin a + \sin b = 2 \sin((a+b)/2) \cos((a-b)/2)$$

15. Geometrie - teoreme in triunghi

$$\text{Teorema sinusurilor: } a/\sin A = b/\sin B = c/\sin C = 2R$$

$$\text{Teorema cosinusurilor: } a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Aria triunghiului: } A = (1/2)ab \sin C = (1/2) \text{ baza} \times h$$

$$\text{Formula lui Heron: } A = \sqrt{s(s-a)(s-b)(s-c)} \quad s = (a+b+c)/2$$

16. Vectori si analiza vectoriala

$$v = (v_x, v_y, v_z); |v| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$u \cdot v = u_x v_x + u_y v_y + u_z v_z = |u||v|\cos(\text{unghi})$$

$$u \times v (\text{produs vectorial}): |uxv| = |u||v|\sin(\text{unghi})$$

$$\text{Dreapta 3D: } (x-x_0)/l = (y-y_0)/m = (z-z_0)/n$$

$$\text{Plan: } ax+by+cz+d=0 \quad (n=(a,b,c) \text{ vectorul normal})$$

$$d(\text{punct, plan}) = |ax_0+by_0+cz_0+d| / \sqrt{a^2+b^2+c^2}$$

Clasa 12 - Exemple rezolvate

Exemple pas cu pas

Limita: $\lim_{x \rightarrow 0} (\sin 2x) / x$

$$= \lim_{x \rightarrow 0} 2 \sin(2x) / (2x) = 2 \times 1 = 2 \quad (\lim \sin t/t = 1)$$

✓

Studiul lui $f(x) = x^2 / (x^2 - 1)$:

Domeniu: $x \neq 1, x \neq -1$

Asimptote verticale: $x=1$ si $x=-1$

Asimptota orizontala: $\lim_{x \rightarrow \infty} x^2 / (x^2 - 1) = 1 \Rightarrow y=1$

$f(x) = -2x / (x^2 - 1)^2$: $f' = 0$ pt $x=0 \Rightarrow$ minim local $f(0)=0$

Integrala $\int_{[0,1]} x \sqrt{1-x^2} dx$:

Substitutie: $t=1-x^2, dt=-2x dx$

$x=0: t=1; x=1: t=0$

$$= -1/2 \int_{[1,0]} \sqrt{t} dt = 1/2 [t^{3/2} / (3/2)]_{-0^+}^1 = 1/2 \times 2/3 = 1/3 \quad \checkmark$$

Cramer: $2x+y=5, x+3y=7$

$$\det(A) = 2 \times 3 - 1 \times 1 = 5$$

$$\det(A_1) = 5 \times 3 - 1 \times 7 = 8 \Rightarrow x = 8/5$$

$$\det(A_2) = 2 \times 7 - 5 \times 1 = 9 \Rightarrow y = 9/5$$

Sol: $(8/5, 9/5)$ ✓

Clasa 12 - Rezumat

Limite & Derivate

- $\lim \sin(x)/x = 1 \ (x \rightarrow 0)$
- $(fg)' = f'g + fg'$; $(f/g)' = (f'g - fg')/g^2$
- L'Hopital: $0/0$ sau $\inf/\inf$

Studiu functie

- $f' > 0$ cresc; $f' < 0$ desc
- $f'' < 0$ concav; $f'' > 0$ convex
- Asimptote: vert, orizont, oblica

Integrale

- $\int x^n = x^{n+1}/(n+1) + C$
- Parti: $\int u \, dv = uv - \int v \, du$
- $\int_a^b f = F(b) - F(a)$

Matrice & Det.

- $\det(2 \times 2) = ad - bc$
- Cramer: $x_i = \det(A_i) / \det(A)$
- $(AB)^T = B^T A^T$

Combinatorica

- $C(n, k) = n! / (k!(n-k)!)$
- Newton: $(a+b)^n = \sum C(n, k) \dots$
- Pascal: $C(n, k) + C(n, k+1) = C(n+1, k+1)$

Probabilitati

- Bayes: $P(H/A) = P(A/H)P(H)/P(A)$
- Binomiala: $E = np$, $D = np(1-p)$
- Normala: $Z = (X - \mu) / \sigma$